Statistical Inference on Global Health Data: A comprehensive analysis of the WHO Life Expectancy Dataset

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The dataset was loaded successfully. We proceeded to do some checks onto the dataset.

```r
# Checking for dimensions
cat("Dimensions:", dim(df), "\n\n")
```

```
## Dimensions: 2864 21
```

```r
# Checking column names
names(df)
```

```
##  [1] "Country"                     "Region"
##  [3] "Year"                        "Infant_deaths"
##  [5] "Under_five_deaths"           "Adult_mortality"
##  [7] "Alcohol_consumption"         "Hepatitis_B"
##  [9] "Measles"                     "BMI"
## [11] "Polio"                       "Diphtheria"
## [13] "Incidents_HIV"               "GDP_per_capita"
## [15] "Population_mln"              "Thinness_ten_nineteen_years"
## [17] "Thinness_five_nine_years"    "Schooling"
## [19] "Economy_status_Developed"    "Economy_status_Developing"
## [21] "Life_expectancy"
```

```r
# Checking data types
str(df)
```

```
## 'data.frame':    2864 obs. of  21 variables:
##  $ Country                    : chr  "Turkiye" "Spain" "India" "Guyana" ...
##  $ Region                     : chr  "Middle East" "European Union" "Asia" "South America" ...
##  $ Year                       : int  2015 2015 2007 2006 2012 2006 2015 2000 2001 2008 ...
##  $ Infant_deaths              : num  11.1 2.7 51.5 32.8 3.4 9.8 6.6 8.7 22 15.3 ...
##  $ Under_five_deaths          : num  13 3.3 67.9 40.5 4.3 11.2 8.2 10.1 26.1 17.8 ...
##  $ Adult_mortality            : num  105.8 57.9 201.1 222.2 58 ...
##  $ Alcohol_consumption        : num  1.32 10.35 1.57 5.68 2.89 ...
##  $ Hepatitis_B                : int  97 97 60 93 97 88 97 88 97 97 ...
##  $ Measles                    : int  65 94 35 74 89 86 97 99 87 92 ...
##  $ BMI                        : num  27.8 26 21.2 25.3 27 26.4 26.2 25.9 27.9 26.5 ...
##  $ Polio                      : int  97 97 67 92 94 89 97 99 97 96 ...
##  $ Diphtheria                 : int  97 97 64 93 94 89 97 99 99 90 ...
##  $ Incidents_HIV              : num  0.08 0.09 0.13 0.79 0.08 0.16 0.08 0.08 0.13 0.43 ...
##  $ GDP_per_capita             : int  11006 25742 1076 4146 33995 9110 9313 8971 3708 2235 ...
##  $ Population_mln             : num  78.53 46.44 1183.21 0.75 7.91 ...
##  $ Thinness_ten_nineteen_years: num  4.9 0.6 27.1 5.7 1.2 2 2.3 2.3 4 2.9 ...
##  $ Thinness_five_nine_years   : num  4.8 0.5 28 5.5 1.1 1.9 2.3 2.3 3.9 3.1 ...
##  $ Schooling                  : num  7.8 9.7 5 7.9 12.8 7.9 12 10.2 9.6 10.9 ...
##  $ Economy_status_Developed   : int  0 1 0 0 1 0 0 1 0 0 ...
##  $ Economy_status_Developing  : int  1 0 1 1 0 1 1 0 1 1 ...
##  $ Life_expectancy            : num  76.5 82.8 65.4 67 81.7 78.2 71.2 71.2 71.9 68.7 ...
```

```r
# Checking for null values
colSums(is.na(df))
```

```
##                     Country                      Region
##                           0                           0
##                        Year               Infant_deaths
##                           0                           0
##           Under_five_deaths             Adult_mortality
##                           0                           0
##         Alcohol_consumption                 Hepatitis_B
```

```
##                         0                              0
##                   Measles                            BMI
##                         0                              0
##                     Polio                     Diphtheria
##                         0                              0
##             Incidents_HIV                 GDP_per_capita
##                         0                              0
##            Population_mln Thinness_ten_nineteen_years
##                         0                              0
##    Thinness_five_nine_years                    Schooling
##                         0                              0
##    Economy_status_Developed   Economy_status_Developing
##                         0                              0
##           Life_expectancy
##                         0
```

We saw that there 2864 rows and 21 columns in the dataset.
There were no missing values in all the variables. In order to be sure, we also proceeded to check if there were rows labeled 'null'.

```r
# Checking if there were values masked as 'null', 'NULL' or " "
cat("Searching for text based masked nulls \n")
```

```
## Searching for text based masked nulls
```

```r
colSums(df == "null" | df == "NULL" | df == "", na.rm = TRUE)
```

```
##                   Country                         Region
##                         0                              0
##                      Year                  Infant_deaths
##                         0                              0
##          Under_five_deaths                Adult_mortality
##                         0                              0
##       Alcohol_consumption                    Hepatitis_B
##                         0                              0
##                   Measles                            BMI
##                         0                              0
##                     Polio                     Diphtheria
##                         0                              0
##             Incidents_HIV                 GDP_per_capita
##                         0                              0
##            Population_mln Thinness_ten_nineteen_years
##                         0                              0
##    Thinness_five_nine_years                    Schooling
##                         0                              0
##    Economy_status_Developed   Economy_status_Developing
##                         0                              0
##           Life_expectancy
##                         0
```

```r
# Checking for suspicious zeros (0) in the health metrics
cat("\n Investigating null values masquerading as 0s \n")
```

```
##
##  Investigating null values masquerading as 0s
```

```r
cat("Zeros in Life Expectancy:", sum(df$Life_expectancy == 0, na.rm = TRUE), "\n" )
```

```
## Zeros in Life Expectancy: 0
```

```r
cat("Zeros in under 5 deaths:", sum(df$Under_five_deaths == 0, na.rm = TRUE), "\n")
```

```
## Zeros in under 5 deaths: 0
```

```r
cat("Zeros in under Adult Mortality:", sum(df$Adult_mortality == 0, na.rm = TRUE), "\n")
```

```
## Zeros in under Adult Mortality: 0
```

```r
cat("Zeros in under alcohol consumption:", sum(df$Alcohol_consumption == 0, na.rm = TRUE),
 ↪  "\n")
```

```
## Zeros in under alcohol consumption: 38
```

```r
cat("Zeros in under BMI:", sum(df$BMI == 0, na.rm = TRUE), "\n")
```

```
## Zeros in under BMI: 0
```

```r
cat("Zeros in under Life Expectancy:", sum(df$Life_expectancy == 0, na.rm = TRUE), "\n")
```

```
## Zeros in under Life Expectancy: 0
```

```r
cat("Zeros in under GDP per capita:", sum(df$GDP_per_capita == 0, na.rm = TRUE), "\n")
```

```
## Zeros in under GDP per capita: 0
```

```r
cat("Zeros in incidents in HIV:", sum(df$Incidents_HIV == 0, na.rm = TRUE), "\n")
```

```
## Zeros in incidents in HIV: 0
```

It was seen that there were no masked values as “null”, ‘null’ or an empty string.
Upon investigating the zeros, there was no suspicious activity of zeros. The selected columns for this check were guided by real world scenarios. This is such as no country could have a GDP of zero or an entire population could have a BMI of zero. We did not check for the various diseases because it is possible for the various developed countries to have incidences where there are no certain diseases because of their well developed health care system which can also be backed by their robust immunization procedures. With regards to alcohol where we obtained 38 zeros, we needed to identify if these countries came from the Muslim majority countries or countries under an Islamic regime of which according to their religious views, consumption of alcohol is prohibited.

```r
# Checking which countries have an alcohol consumption of zero
unique(df$Country[df$Alcohol_consumption == 0])
```

```
## [1] "Saudi Arabia" "Kuwait"       "Somalia"      "Bangladesh"   "Mauritania"
## [6] "Afghanistan"  "Libya"
```

The Muslim majority hypothesis proved to be correct as the countries with 0 alcohol consumption came from Islamic nations.
We needed to check if the highly-developed countries indeed had no vaccine preventable diseases. We also proceeded to investigate if presence of zeros meant that no immunization coverage was done or it meant that there were no such cases.

```r
# Checking if it was immunization coverage or case count
cat("Summary of disease columns \n")
```

```
## Summary of disease columns
```

```r
summary(df[ , c("Measles", "Polio", "Diphtheria")])
```

```
##     Measles          Polio         Diphtheria
##  Min.   :10.00   Min.   : 8.0   Min.   :16.00
##  1st Qu.:64.00   1st Qu.:81.0   1st Qu.:81.00
```

```
##  Median :83.00   Median :93.0   Median :93.00
##  Mean   :77.34   Mean   :86.5   Mean   :86.27
##  3rd Qu.:93.00   3rd Qu.:97.0   3rd Qu.:97.00
##  Max.   :99.00   Max.   :99.0   Max.   :99.00
```

An inspection of these diseases proved that there was a strict maximum threshold of 99. For actual cases, this value would scale to higher figures such as tens of thousands. This evidently proved that these values were actual percentage coverage rather than actual case counts.
We needed to do an investigation of these percentages. This was to check whether the percentages meant 99% covered or not covered. This was done by the use of a correlation. If it meant covered, there would be an inverse relationship between the disease and `Infant_deaths` or `Under_five_deaths`.

```r
cat("The correlation between the Measles disease (coverage) and death of
    children under 5 years is", cor(df$Measles, df$Under_five_deaths, use = 'complete.obs'),
 ↪  "\n\n")
```

```
## The correlation between the Measles disease (coverage) and death of
##     children under 5 years is -0.5129717
```

```r
cat("The correlation between the Polio disease (coverage) and death of
    infants is", cor(df$Polio, df$Infant_deaths, use = 'complete.obs'), "\n\n")
```

```
## The correlation between the Polio disease (coverage) and death of
##     infants is -0.7407905
```

This showed an inverse relationship between the diseases and the child mortality. This meant that the values exhibited by the diseases meant the extensiveness of the vaccinations. A 99% coverage meant that most parts of this particular country was vaccinated against the specific disease.

The project's next step was to check if indeed the high immunization coverage came from the highly developed countries. This would validate our hypothesis of developed countries having better healthcare. We began by investigating the robustness of `Economy_status_Developing` and `Economy_status_Developed` features to ensure that they were mutually exclusive.

```r
# Investigating that the two economy columns do not add up to 2. Mutual
# exclusivity is needed
economy_check <- df$Economy_status_Developing + df$Economy_status_Developed

# Summary of the calculation
cat("Summary of the economy checks \n")
```

```
## Summary of the economy checks
```

```r
summary(economy_check)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##       1       1       1       1       1       1
```

```r
# Tabular form of these two columns
cat("\n Cross validation table of the two economy features \n")
```

```
##
##  Cross validation table of the two economy features
```

```r
table(Developed = df$Economy_status_Developed, Developing = df$Economy_status_Developing)
```

```
##          Developing
## Developed    0    1
```

```
##          0    0 2272
##          1  592    0
```

We saw that the two columns were indeed mutually exclusive as they both did not add up to 2. The cross validation table backs this up whereby there are no individual countries where they are both developed and developing. This is given by the `0`. Additionally, the table also informs us that there are `2272` records from developing countries and there are `592` records from developed countries.

We now proceeded to see if the high vaccination coverage came from developed or developing countries.

```r
# Comparing the coverage between the two economy statuses
cat("Average coverage: Developing (0) vs Developed (1) \n")
```

```
## Average coverage: Developing (0) vs Developed (1)
```

```r
aggregate(cbind(Measles, Polio, Diphtheria) ~ Economy_status_Developed, data = df, FUN = mean,
↪  na.rm = TRUE)
```

```
##   Economy_status_Developed  Measles    Polio Diphtheria
## 1                        0 74.50044 84.31954   83.97711
## 2                        1 88.26182 94.86655   95.07770
```

```r
# Checking which economy status holds 0% coverage record for measles
cat("Economy status for countries with 0% Measles coverage \n")
```

```
## Economy status for countries with 0% Measles coverage
```

```r
table(Measles_zero = df$Measles == 0, Developed = df$Economy_status_Developed)
```

```
##             Developed
## Measles_zero    0    1
##        FALSE 2272  592
```

We saw that developed countries had a better coverage of vaccination as compared to developing countries. For instance, Measles had a 88.26% coverage in developed countries while it had a 74.5% coverage in the developing countries. This trend was seen in Polio and Diphtheria.
The table also showed that there was no instance of any country where it had a 0% coverage of Measles. Every single row returned `FALSE` thus proving coverage across all countries.
We then proceeded to conduct descriptive statistics.

```r
# Isolating the numeric columns as well as omitting variables which
# would be less meaningful to have summary statistics
cols_to_exclude <- c("Year", "Economy_status_Developed", "Economy_status_Developing")
num_df <- df[ , sapply(df, is.numeric)]
num_df <- num_df [, !(names(num_df) %in% cols_to_exclude)]

# Calculating the statistics
stats <- function(x) {
  x <- na.omit(x)
  c(
    Mean = mean(x),
    Median = median(x),
    Variance = var(x),
    SD = sd(x),
    Range = max(x) - min(x),
    IQR = IQR(x)
  )
}
```

```r
stats_table <- t(sapply (num_df, stats))

cat ("Descriptive statistics \n")
```

```
## Descriptive statistics
```

```r
round(stats_table, 2)
```

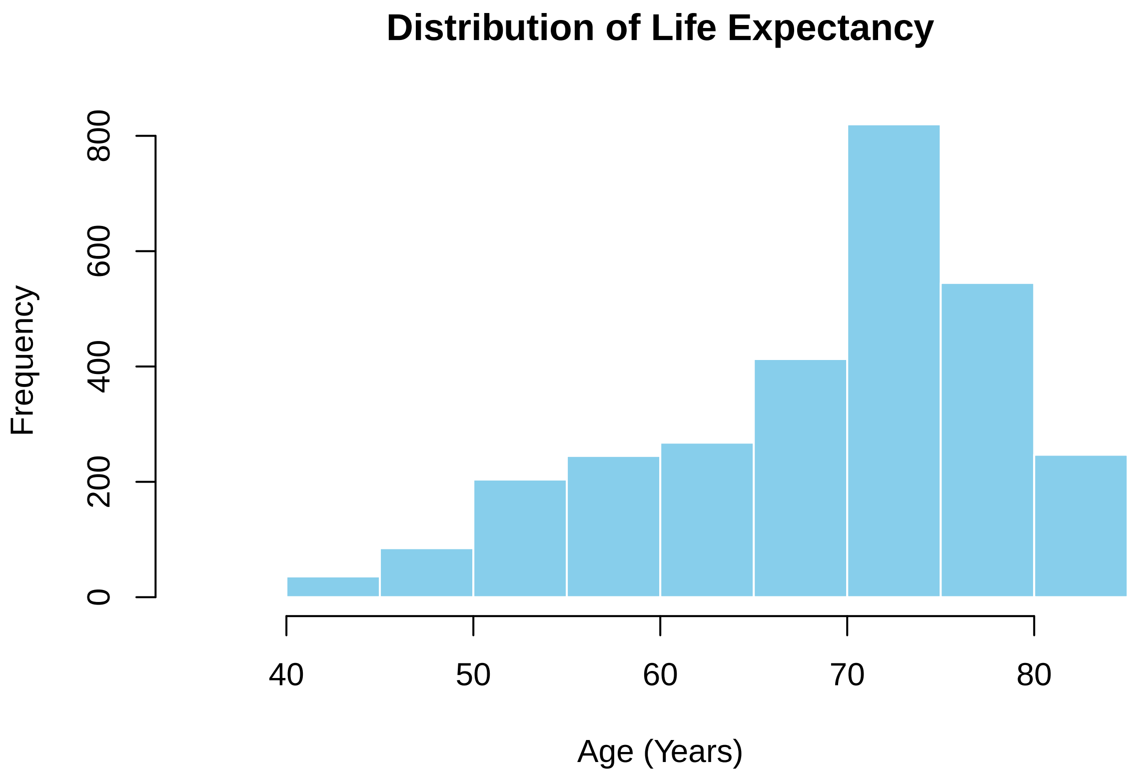
```
##                                Mean  Median     Variance       SD     Range
## Infant_deaths                 30.36   19.60       758.35    27.54    136.30
## Under_five_deaths             42.94   23.10      1986.48    44.57    222.60
## Adult_mortality              192.25  163.84     13204.37   114.91    669.98
## Alcohol_consumption            4.82    4.02        15.86     3.98     17.87
## Hepatitis_B                   84.29   89.00       255.86    16.00     87.00
## Measles                       77.34   83.00       348.18    18.66     89.00
## BMI                           25.03   25.50         4.81     2.19     12.30
## Polio                         86.50   93.00       227.42    15.08     91.00
## Diphtheria                    86.27   93.00       241.31    15.53     83.00
## Incidents_HIV                  0.89    0.15         5.67     2.38     21.67
## GDP_per_capita             11540.92 4217.00 286787076.14 16934.79 112270.00
## Population_mln                36.68    7.85     18628.39   136.49   1379.78
## Thinness_ten_nineteen_years    4.87    3.30        19.70     4.44     27.60
## Thinness_five_nine_years       4.90    3.40        20.48     4.53     28.50
## Schooling                      7.63    7.80        10.06     3.17     13.00
## Life_expectancy               68.86   71.40        88.47     9.41     44.40
##                                 IQR
## Infant_deaths                 39.25
## Under_five_deaths             56.33
## Adult_mortality              139.88
## Alcohol_consumption            6.58
## Hepatitis_B                   18.00
## Measles                       29.00
## BMI                            3.20
## Polio                         16.00
## Diphtheria                    16.00
## Incidents_HIV                  0.38
## GDP_per_capita             11141.25
## Population_mln                21.59
## Thinness_ten_nineteen_years    5.60
## Thinness_five_nine_years       5.70
## Schooling                      5.20
## Life_expectancy               12.70
```

From the summary statistics above, we saw no major outliers or anomalies that would have warranted an immediate action. Although, there was possible skewness `GDP_per_capita` whereby it had a median of *4217.00* and a mean of *11540.92*. A group of high income individuals would cause this. Such trends were also witnessed in `Population_mln` and `Incidents_HIV`, where the former could be explained by countries having a high population. With that, we proceeded to generate visualizations which would help us see missed information.

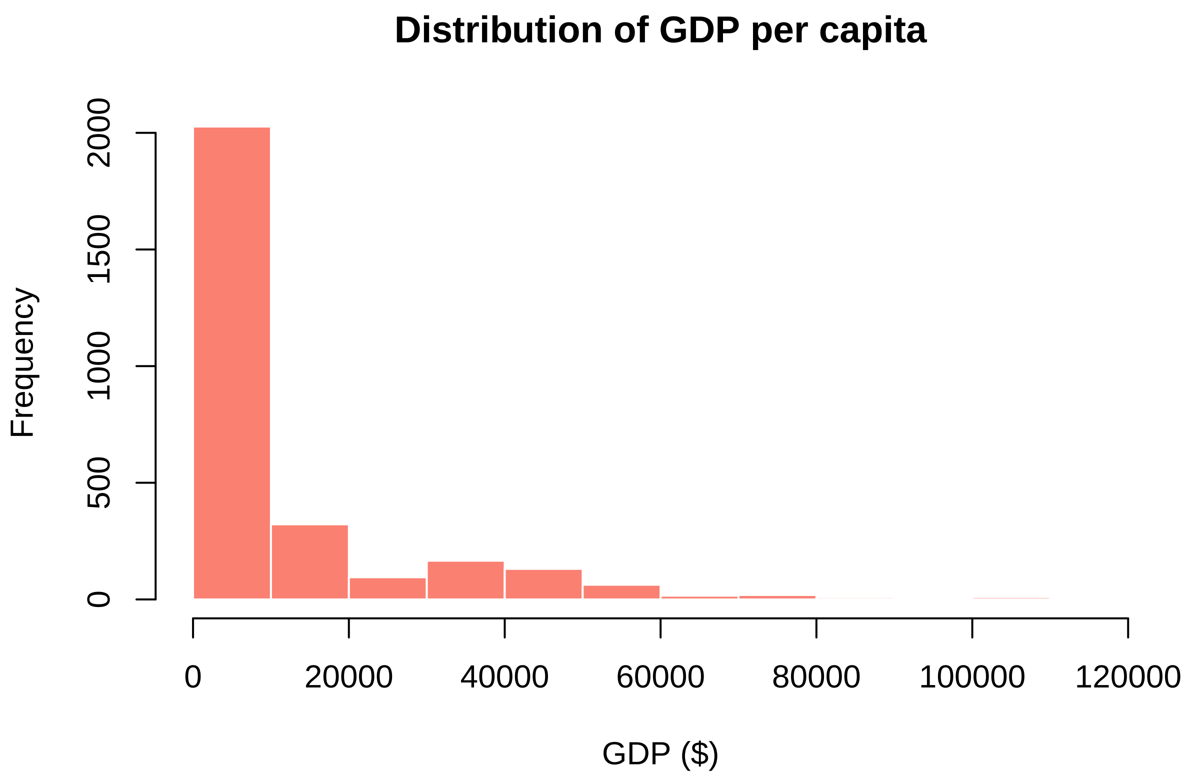
We began by histograms:

```r
# Life Expectancy
hist(df$Life_expectancy,
     main = "Distribution of Life Expectancy",
     xlab = "Age (Years)",
```

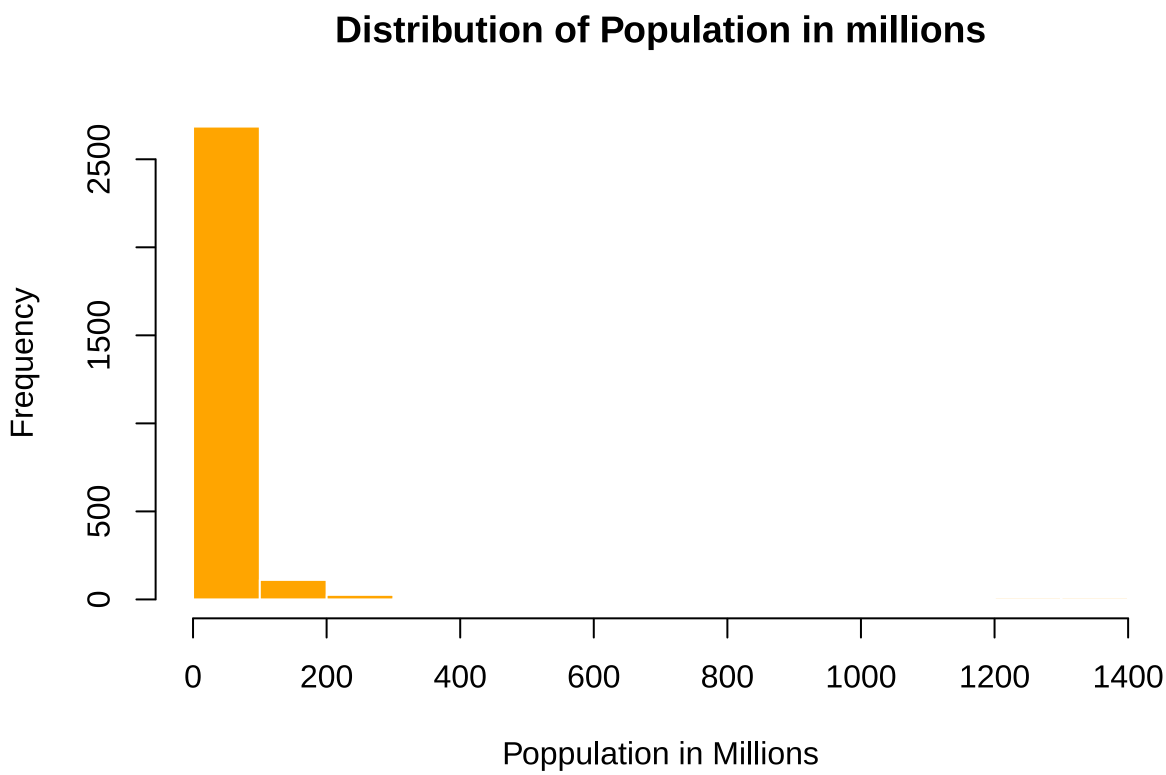
```r
     col = "skyblue",
     border = "white")
```



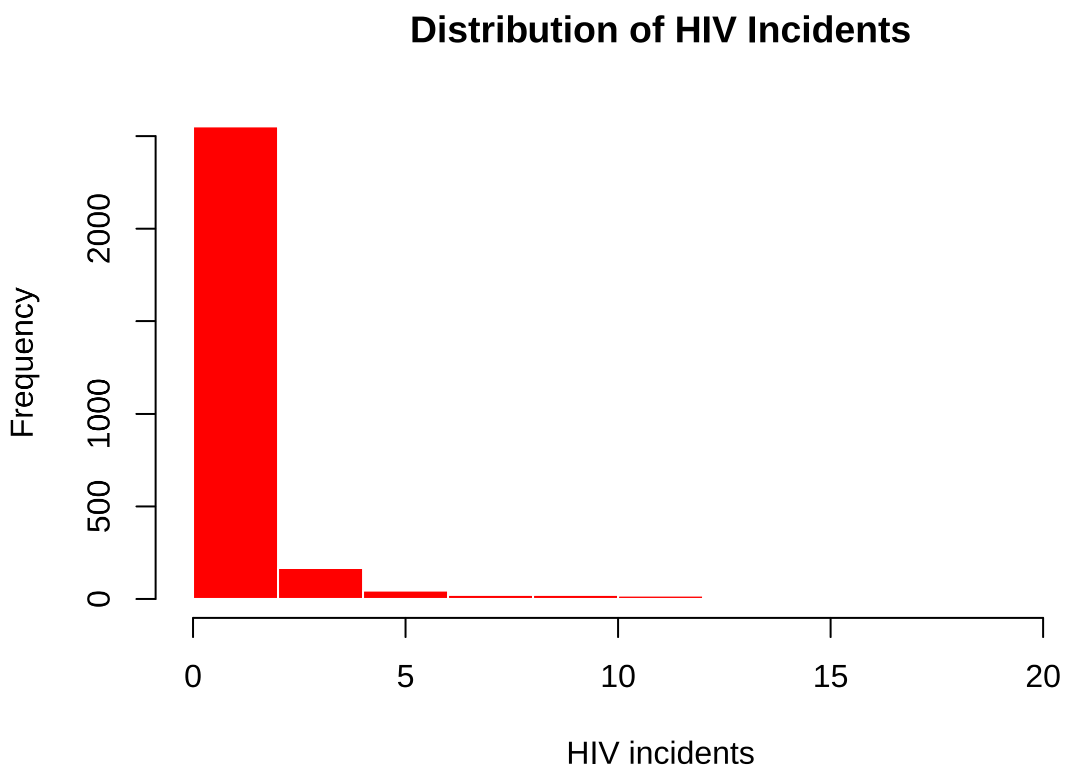

```r
# GDP Per Capita
hist(df$GDP_per_capita,
     main = "Distribution of GDP per capita",
     xlab = "GDP ($)",
     col = "salmon",
     border = "white")
```




```r
# Population in Million
hist(df$Population_mln,
     main = "Distribution of Population in millions",
     xlab = "Poppulation in Millions",
     col = "orange",
     border = "white")
```



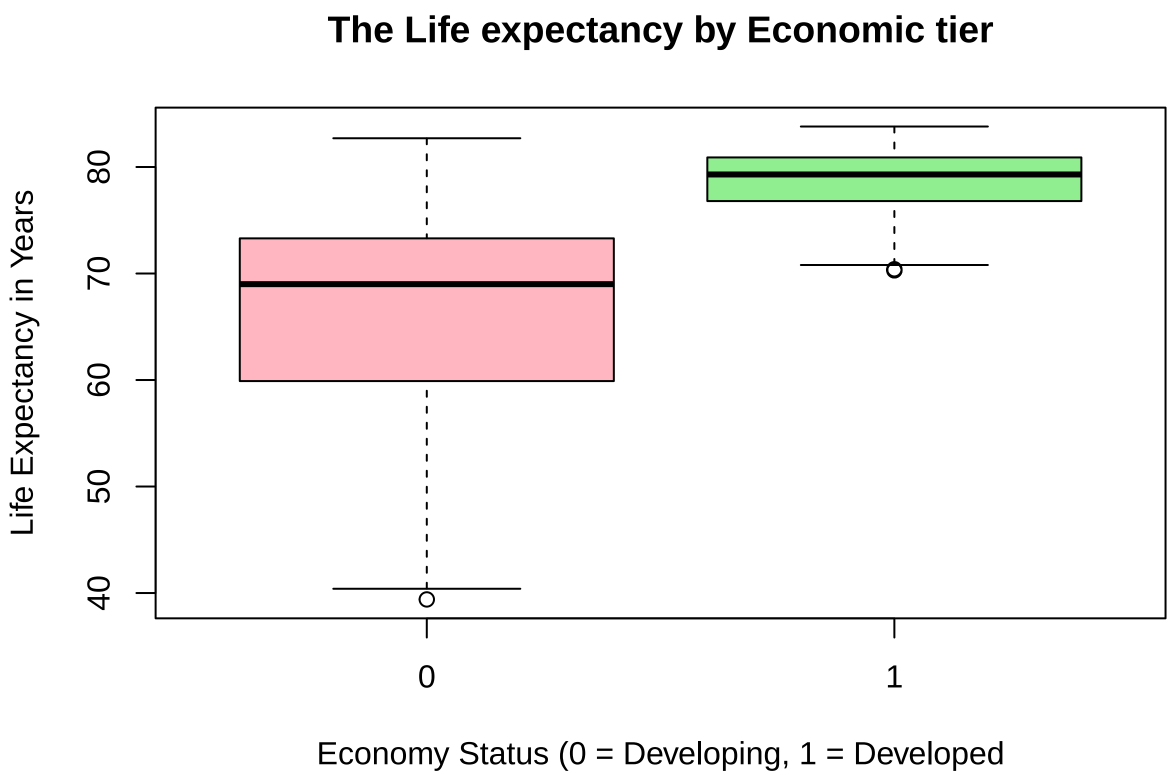

```r
# Incidents in HIV
hist(df$Incidents_HIV,
     main = "Distribution of HIV Incidents",
     xlab = "HIV incidents",
     col = "red",
     border = "white")
```



As previously explained by the numbers, the histograms revealed that the `GDP_per_capita`, `Population_mln` and `Incidents_HIV` were positively skewed.
The `Life_Expectancy` histogram was negatively skewed as expected, peaking at the ≈ 75 age mark.

We proceeded to construct boxplots.

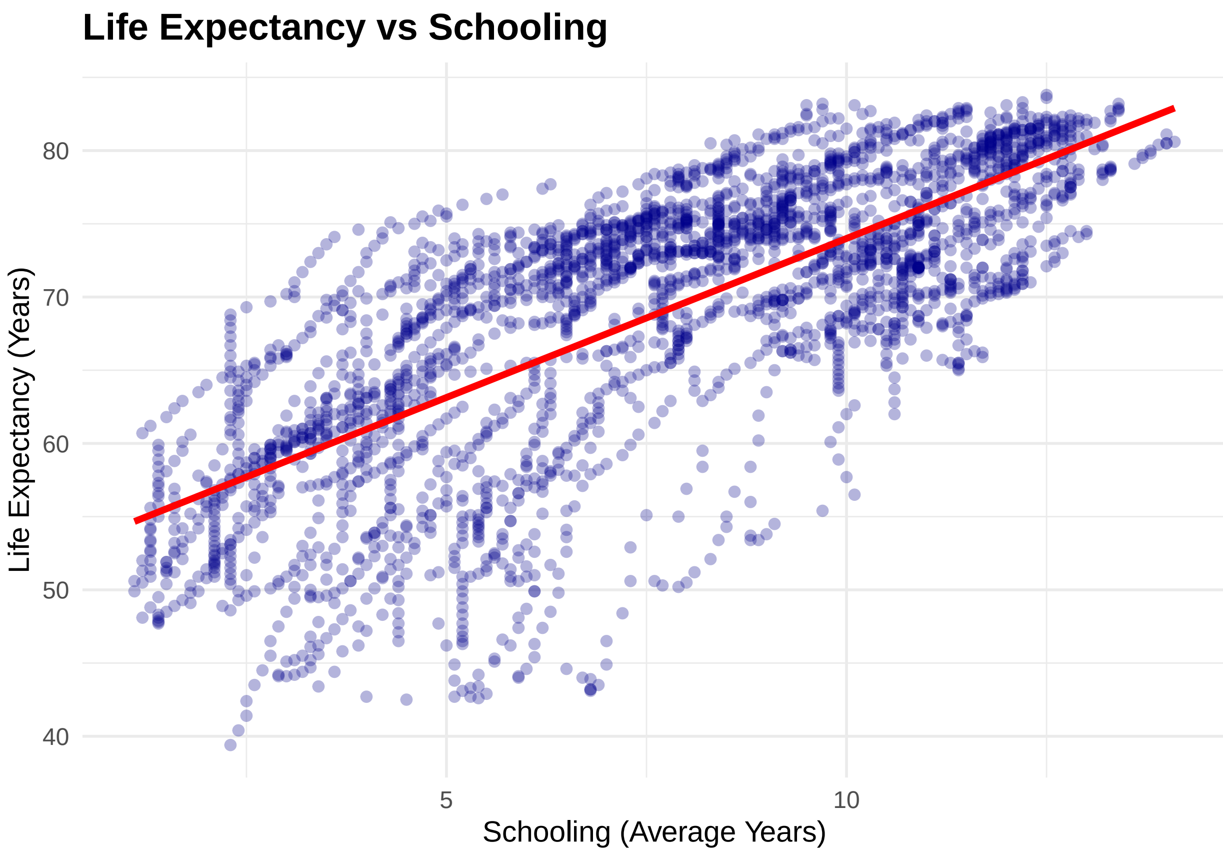
```
# Life expectancy with economy status developed
boxplot(Life_expectancy ~ Economy_status_Developed,
        data = df,
        main = "The Life expectancy by Economic tier",
        xlab = "Economy Status (0 = Developing, 1 = Developed",
        ylab = "Life Expectancy in Years",
        col = c("lightpink", "lightgreen"))
```



From the boxplot, we saw that developed countries have a higher life expectancy than developing countries. This can as well be backed up by the vaccination coverage we identified previously. Developed countries have a higher coverage than developing countries.

We proceeded to construct a scatterplot of `Life_Expectancy` against `schooling`. We desired to visualize their relationship.

```
ggplot(df, aes(x = Schooling, y = Life_expectancy)) +
  geom_point(color = "darkblue", alpha = 0.3) +
  geom_smooth(method = "lm", color = "red", se = FALSE, linewidth = 1.2) +
  labs(title = "Life Expectancy vs Schooling", x = "Schooling (Average Years)", y = "Life
   ↪  Expectancy (Years)") +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14))
```

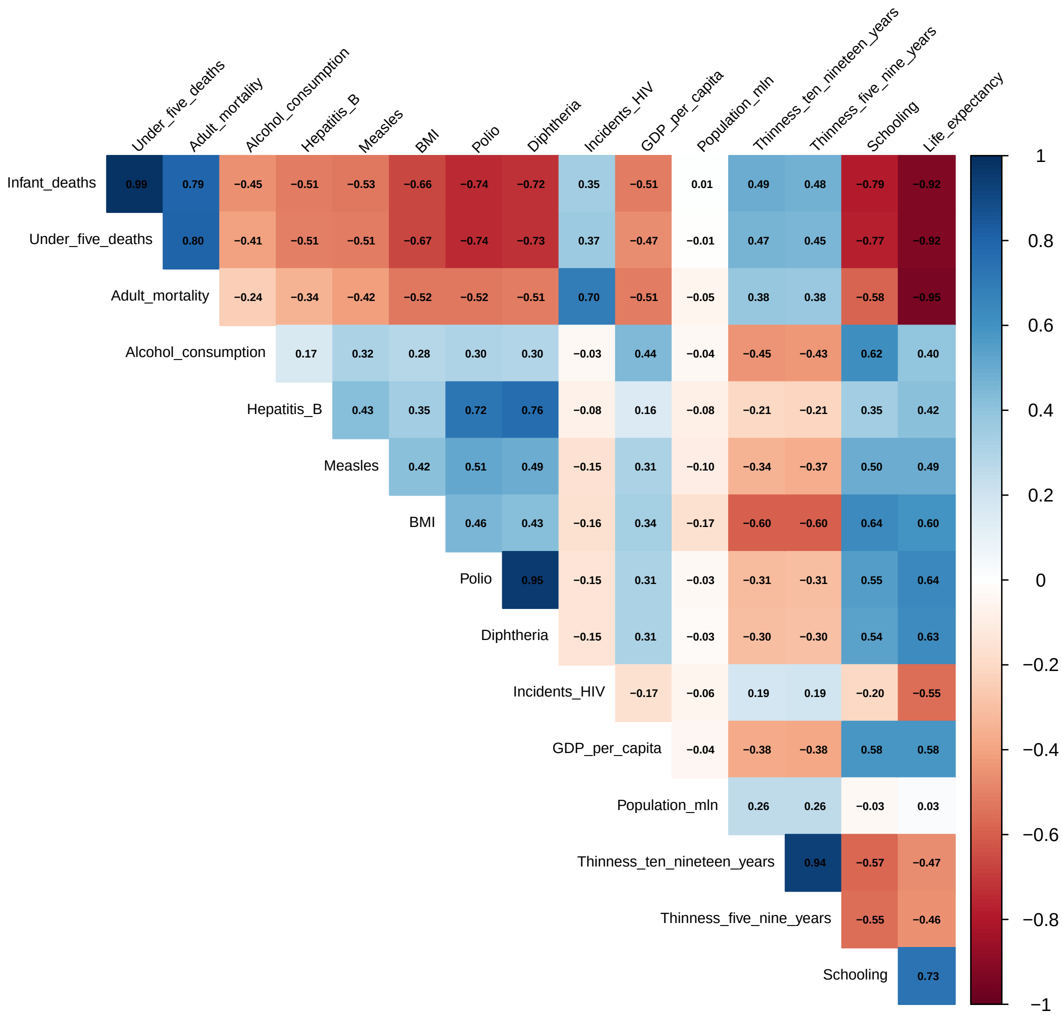


From the scatterplot, we saw that there was a positive relationship between `Life_expectancy` and `Schooling`. The more learned a country is, the higher the life expectancy. This is due to advanced education system yielding better research in how to make better health decisions as well as better standards of living.

We proceeded to have a correlation heatmap.

```
# For only numeric columns
cor_matrix <- cor(num_df, use = "complete.obs")

corrplot(cor_matrix,
        method = "color",
        type = "upper",
        addCoef.col = "black",
        number.cex = 0.5,
        tl.col = "black",
        tl.cex = 0.7,
        tl.srt = 45,
        diag = FALSE)
```



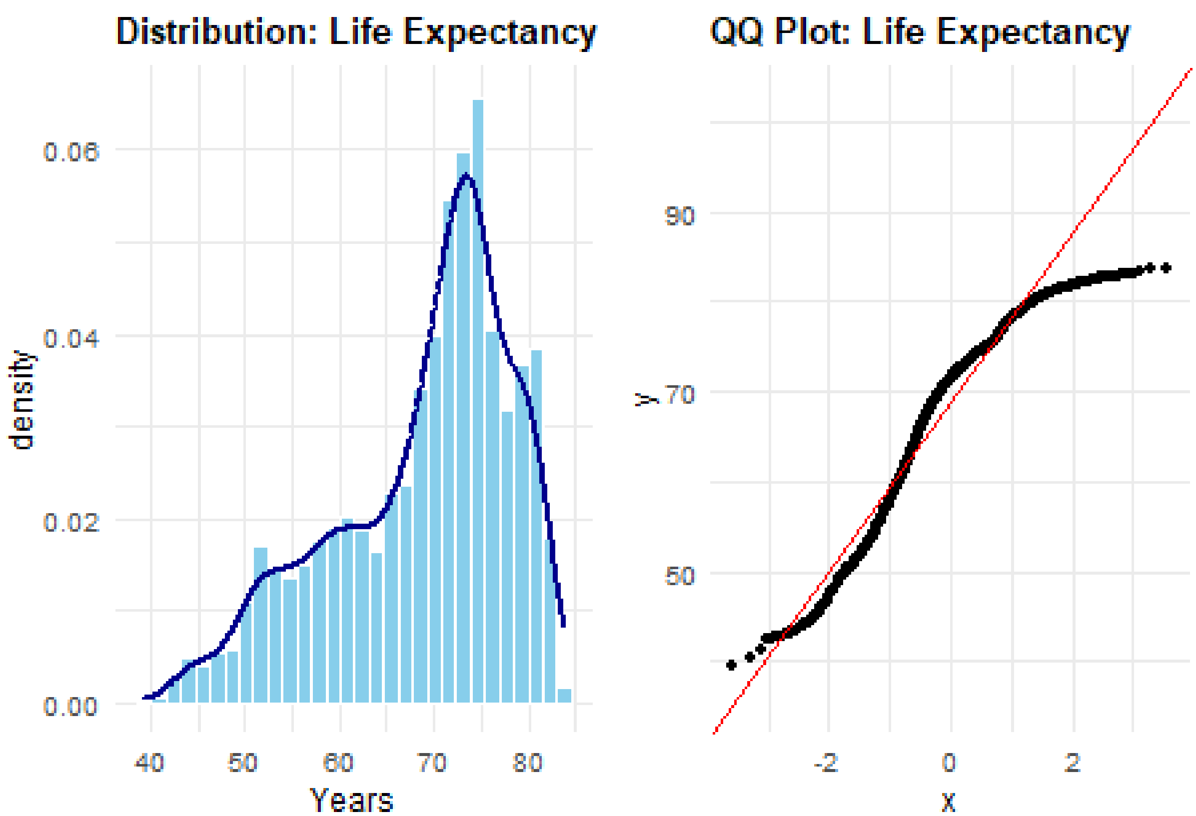
The heatmap validated the scatterplot as there there was a sufficiently high degree of correlation (`0.73`) between `Schooling` and `Life_expectancy`. We also saw a moderate positive relationship between `Life_expectancy` and `GDP_per_capita` as it was valued at `0.58`.
There was a very strong negative relationship (`0.95`) between `Life_expectancy` and `Adult_mortality`. This inverse relationship can be explained as the life expectancy increases, it results to a decrease in the rates of adult mortality.
We also saw evidence of multicollinearity amongst certain variables such as `Polio` and `Diphtheria`, `Infant_deaths` and `Under_five_deaths`, and `Thinness_ten_nineteen_years` and `Thinness_five_nine_years`. These variables almost had a perfect relationship. Subsequent steps would handle this to prevent our tests from crushing.

We proceeded to address the distributional assumptions. We selected 3 variables for this. We picked `Life_expectancy` as it was our dependent variable. Previously, we had observed that `GDP_per_capita` was skewed. This was also selected to see how it violated normality. We also selected a more symmetric variable which was `schooling` to serve as a comparison metric to the skewed variable.

```r
# Density plot + Histogram (Saved as p1)
p1 <- ggplot(df, aes(x = Life_expectancy)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30, fill = "skyblue", color = "white") +
  geom_density(color = "darkblue", linewidth = 1) +
  labs(title = "Distribution: Life Expectancy", x = "Years")

# QQ Plot (Saved as p2)
p2 <- ggplot(df, aes(sample = Life_expectancy)) +
  stat_qq() + stat_qq_line(color = "red") +
  labs(title = "QQ Plot: Life Expectancy")

# Combine plots side-by-side using patchwork
p1 + p2
```




```r
# Shapiro-Wilk
shapiro.test(df$Life_expectancy)
```

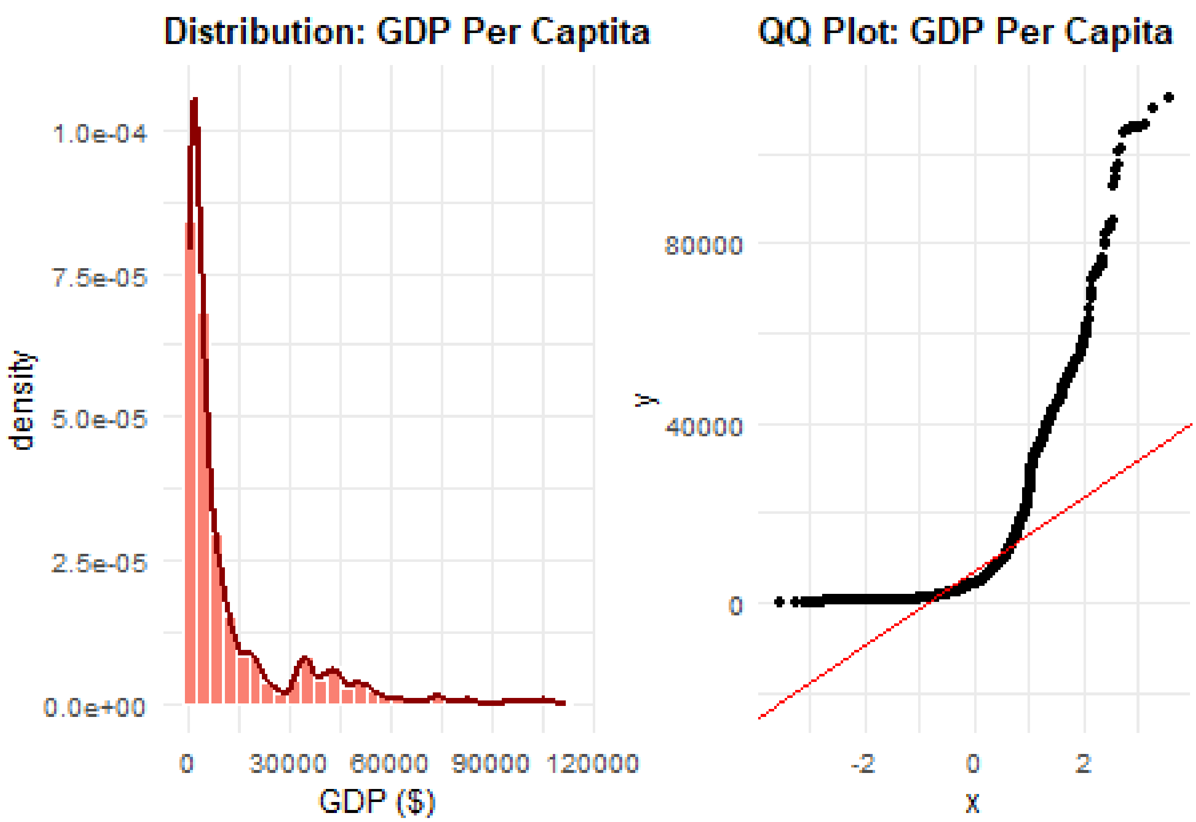
```
##
##  Shapiro-Wilk normality test
##
## data:  df$Life_expectancy
## W = 0.93497, p-value < 2.2e-16
```

The Histogram & Density plots showed that the life expectancy was negatively skewed with a peak at the ≈ 75 mark. The QQ Plot visualized the violation of normality as the plot had a heavy tail, deviating from the red line. Finally the shapiro-wilk test had a very low p-value thus indicating that normality is indeed violated. This would warrant a transformation before formally engaging in the various tests.
The next assessment was on the `GDP_per_capita`.

```r
# Density plot + Histogram
p1 <- ggplot(df, aes(x = GDP_per_capita)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30, fill = "salmon", color = "white") +
  geom_density(color = "darkred", linewidth = 1) +
  labs(title = "Distribution: GDP Per Captita", x = "GDP ($)")

# QQ Plot
p2 <- ggplot(df, aes(sample = GDP_per_capita)) +
  stat_qq() + stat_qq_line(color = "red") +
  labs(title = "QQ Plot: GDP Per Capita")

# Combine plots
p1 + p2
```




```r
# Shapiro-Wilk
shapiro.test(df$GDP_per_capita)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  df$GDP_per_capita
## W = 0.67054, p-value < 2.2e-16
```

From the histogram and density plot, the `GDP_per_capita` was positively skewed. This was also evidenced by the QQ Plot which had much heavy tails as the points did not lie on the red line. The shapiro-wilk test gave a p-value close to zero indicative of the violation of normality.
We finally plotted for `schooling` variable.